

Homework (Carolina Reaper)

- Q1.** A rhythm pattern has 3^4 possible variations. How many variations are there?
(A) 80
(B) 81
(C) 82
(D) 83
(E) 84
- Q2.** A canvas has an area of 2^6 square feet. What is its area in square feet?
(A) 60
(B) 62
(C) 64
(D) 66
(E) 68
- Q3.** The frequency of a sound wave is $2^3 \cdot 5^2$ Hz. What is its frequency?
(A) 100 Hz
(B) 120 Hz
(C) 140 Hz
(D) 160 Hz
(E) 200 Hz
- Q4.** A musical composition has 5^3 possible rhythm patterns. How many patterns are there?
(A) 100
(B) 120
(C) 125
(D) 130
- (E) 140
- Q5.** The volume of a speaker is $3^2 \cdot 2^4$ cubic feet. What is its volume?
(A) 144 cubic feet
(B) 150 cubic feet
(C) 156 cubic feet
(D) 160 cubic feet
(E) 170 cubic feet
- Q6.** A painter has $2^5 \cdot 3$ different brush sizes. How many brush sizes are there?
(A) 90
(B) 96
(C) 100
(D) 110
(E) 120
- Q7.** The length of a canvas is $2^2 \cdot 3^2$ feet. What is its length?
(A) 30 feet
(B) 32 feet
(C) 36 feet
(D) 40 feet
(E) 45 feet
- Q8.** The area of a mural is $5^2 \cdot 2^3$ square feet. What is its area?
(A) 400 square feet
(B) 450 square feet
(C) 500 square feet
(D) 550 square feet
(E) 600 square feet

Open Ended Questions (FRQ)

Q9. Simplify the expression: $\frac{2^5 \cdot 3^2}{2^3 \cdot 3}$

Q10. A sound wave has a frequency of $2^4 \cdot 5$ Hz. Simplify the expression and find the frequency.

Chef's Tips

- Check calculations carefully
- Use exponent rules to simplify expressions
- Ensure students provide clear and concise answers